

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 40
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 40
samples_per_sec: 25
forecast_horizon_sec: 2
forecast_steps: 50
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-11 09:02:14

Training Start: 2025-05-11_01-13-48

Training End: 2025-05-11_01-36-08

Training Duration: 22.34 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 68293.0621

MSE: 7462927128.8205

RMSE: 86388.2349

R2 Score: 0.7218

Multi-Step Forecast:

Forecast Steps: 50

MAE: 0.0341

MSE: 0.0018

RMSE: 0.0424

R2 Score: 0.8113

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 25,646 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

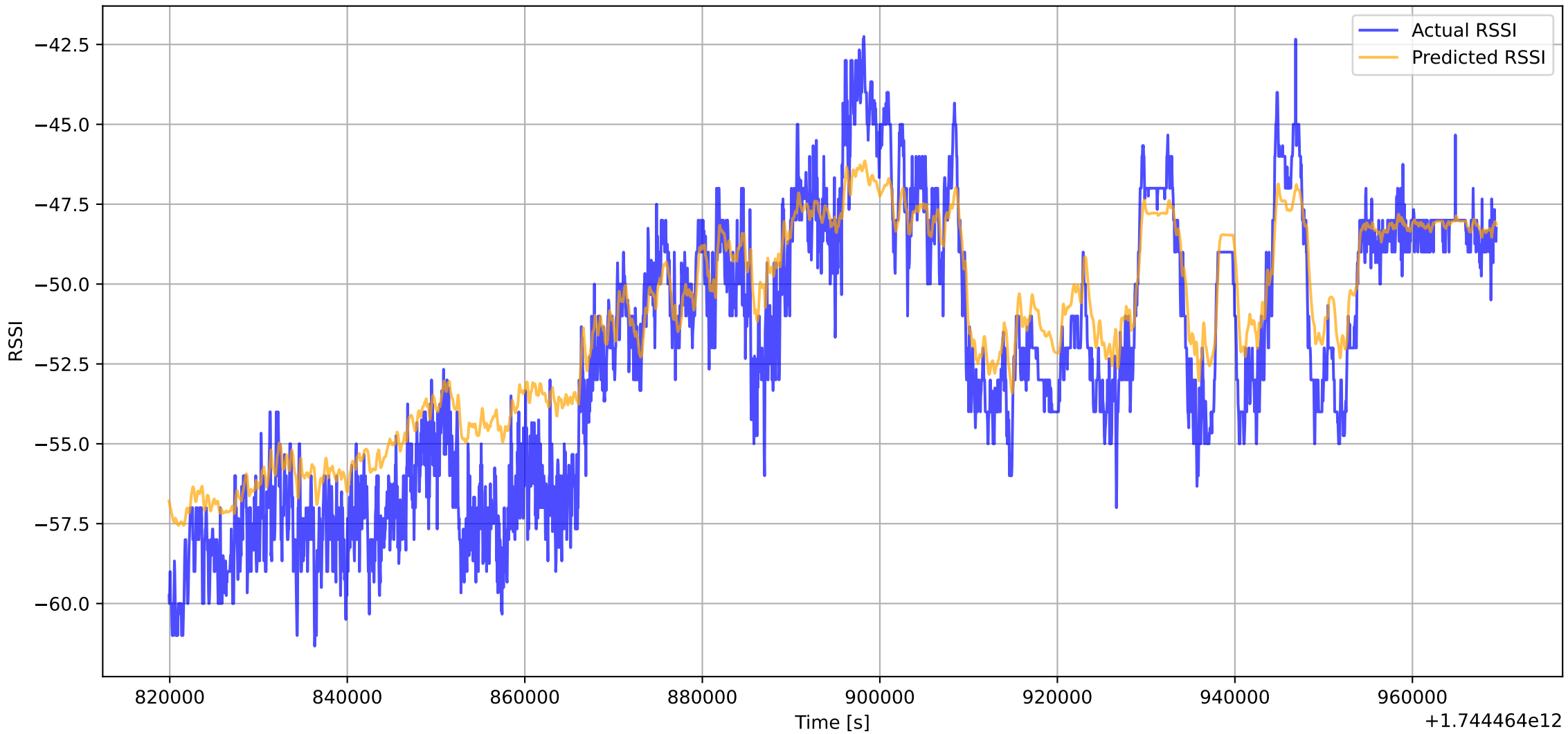
1744397085760
1744397085801
1744397085841
1744397086014
1744397086041

...

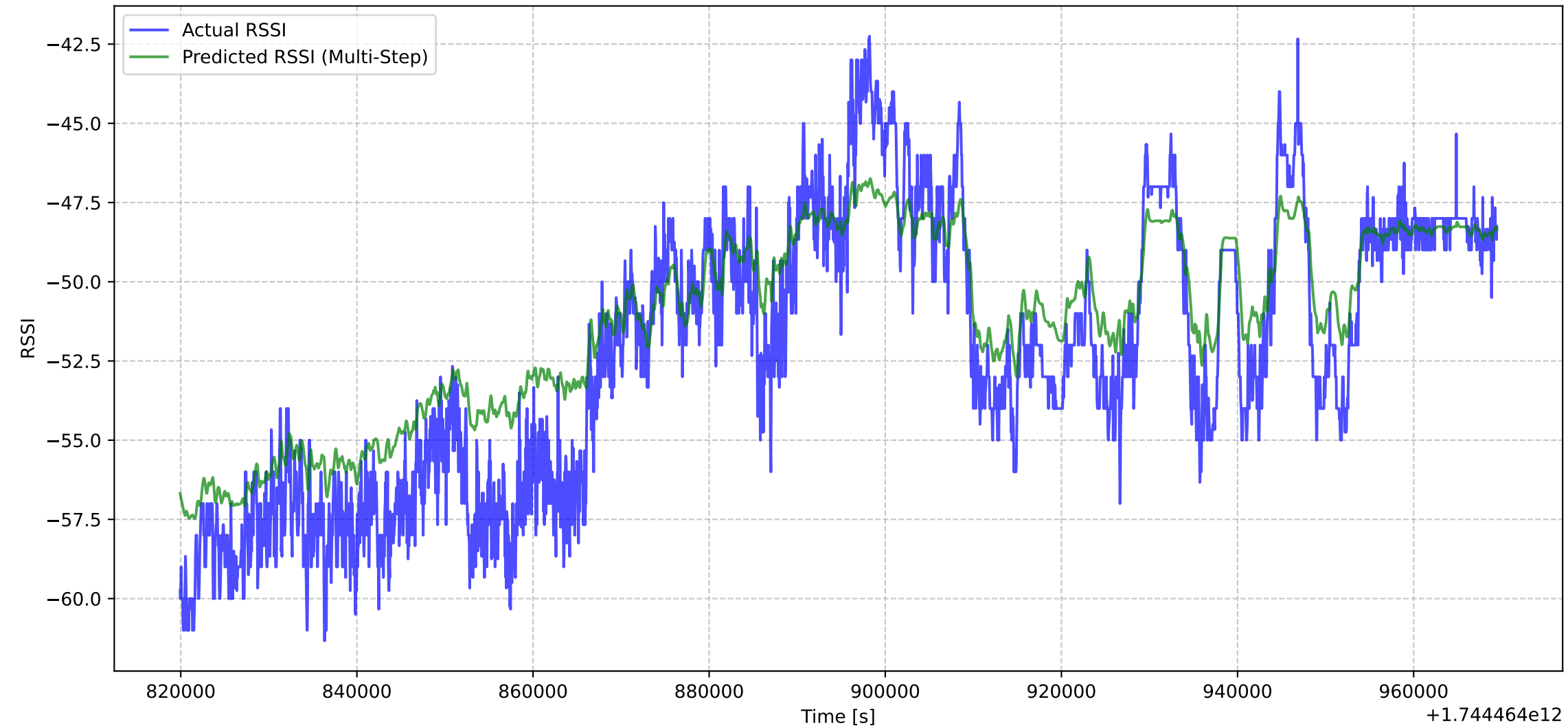
Last 5 timestamps AFTER:

1744464971245
1744464971281
1744464971329
1744464971366
1744464971402

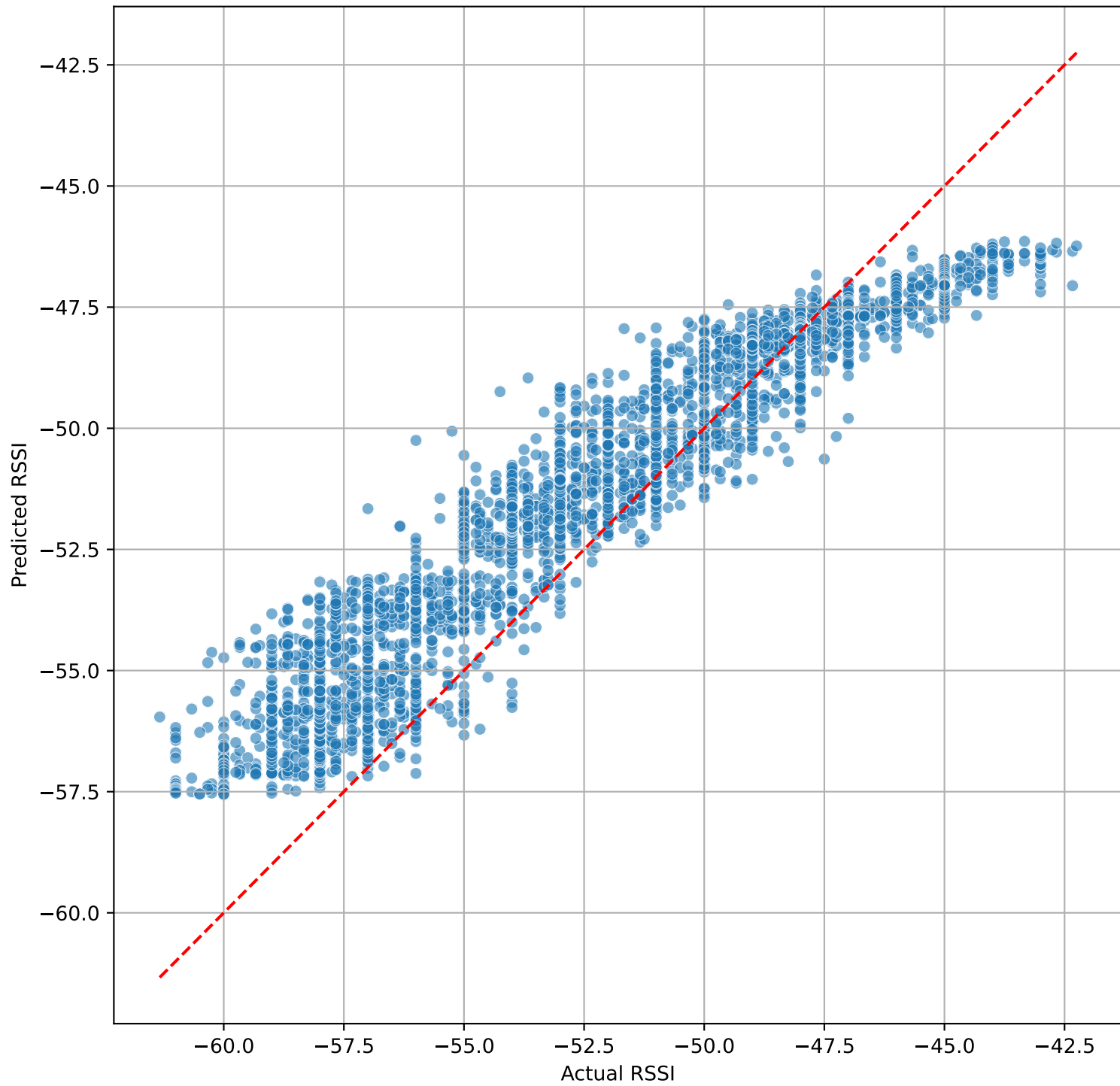
Actual vs Predicted RSSI (First Step)



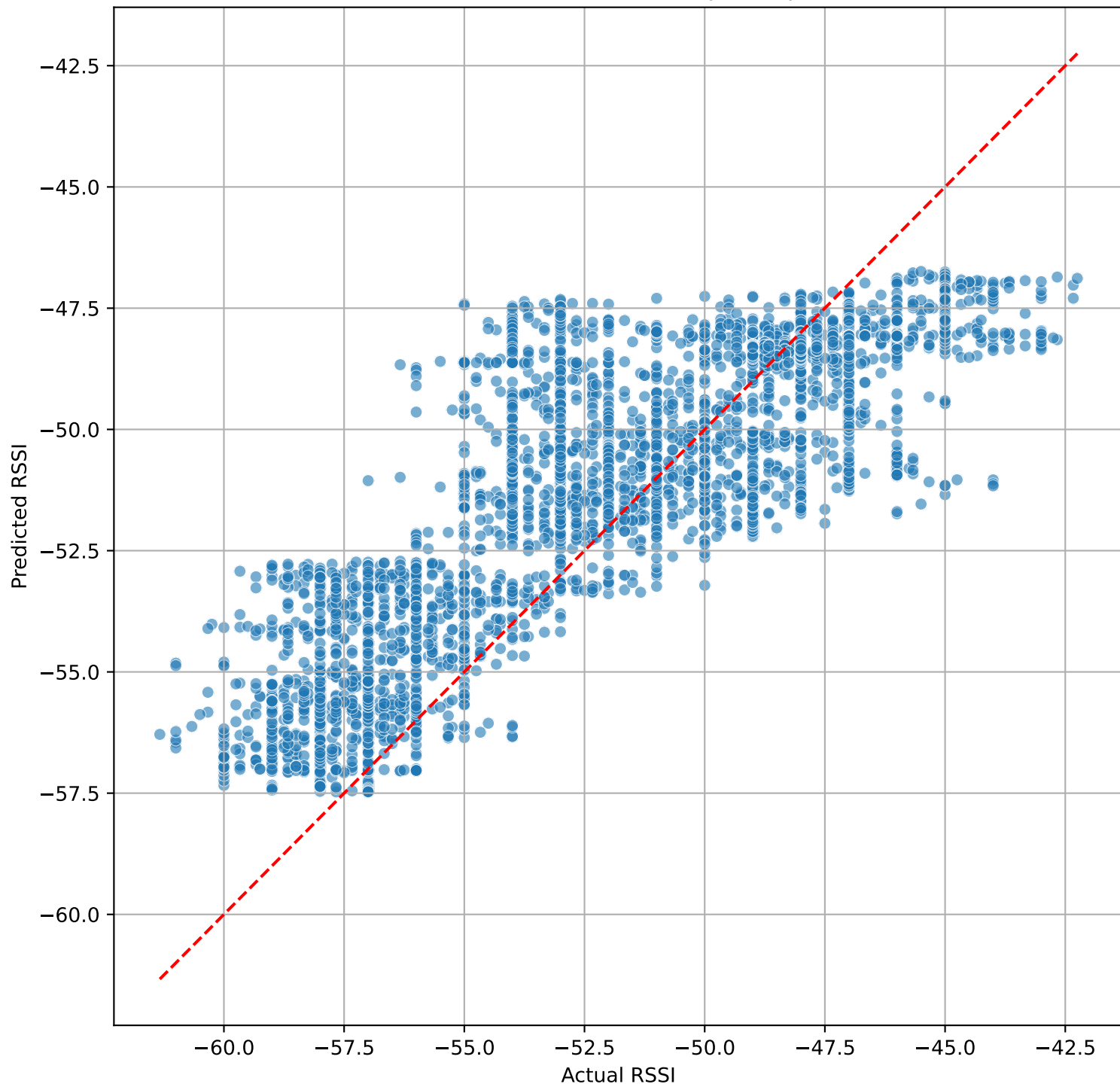
Actual vs Predicted RSSI (Multi-Step (step 50 $\approx$ 2.0 s))



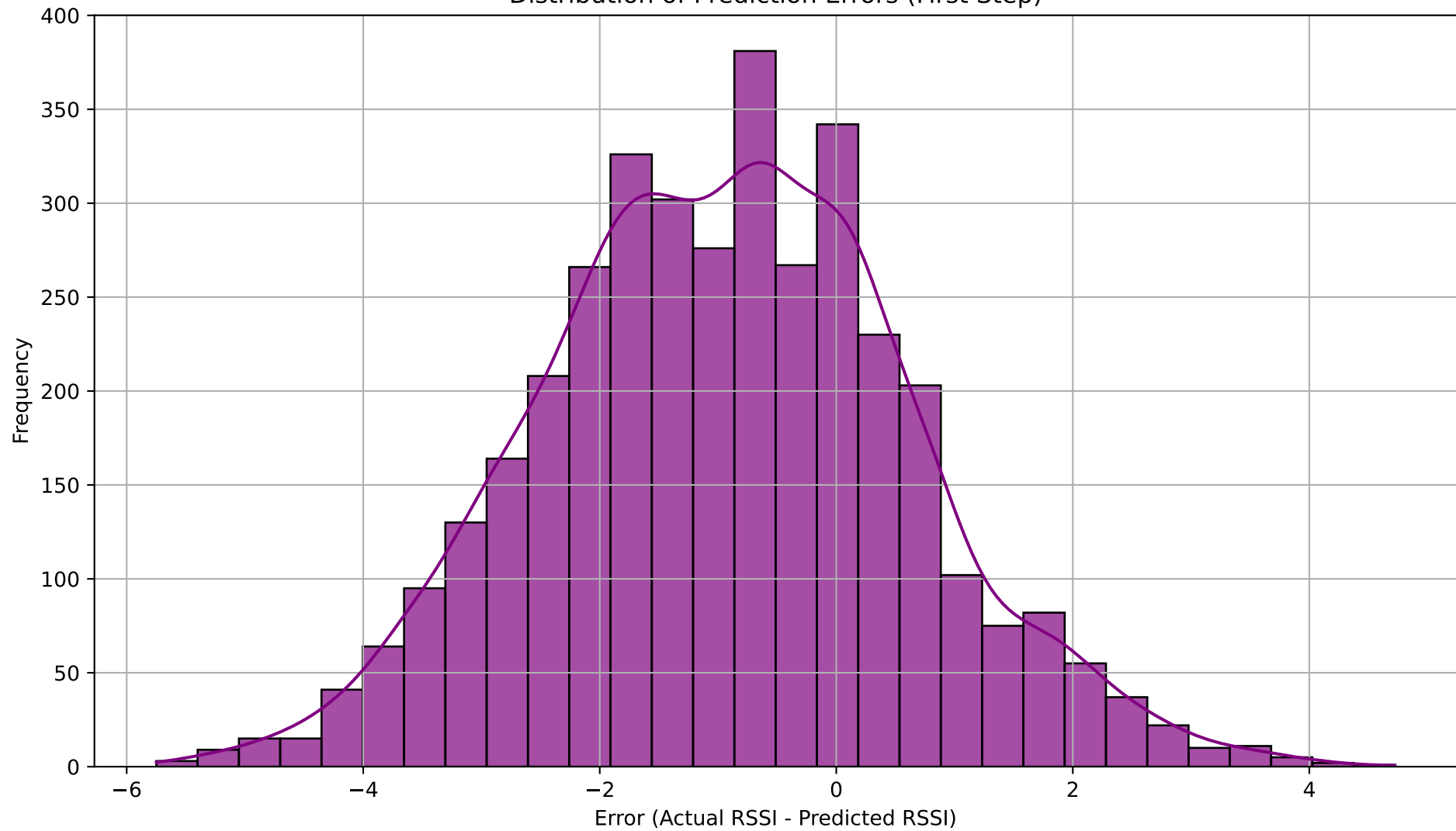
Actual vs Predicted RSSI (First Step)



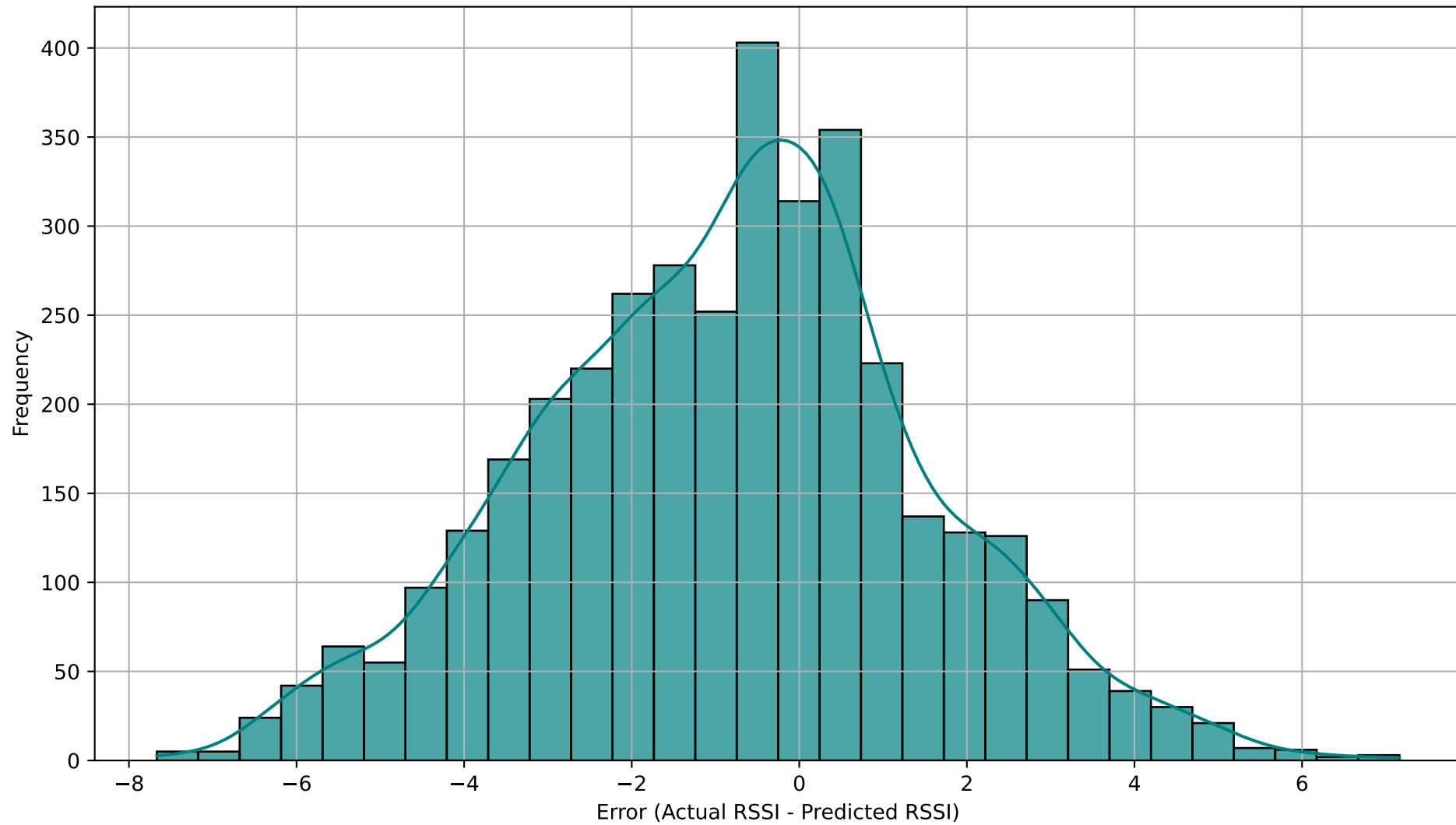
Actual vs Predicted RSSI (Last Step - step 50 $\approx$ 2.0 s)



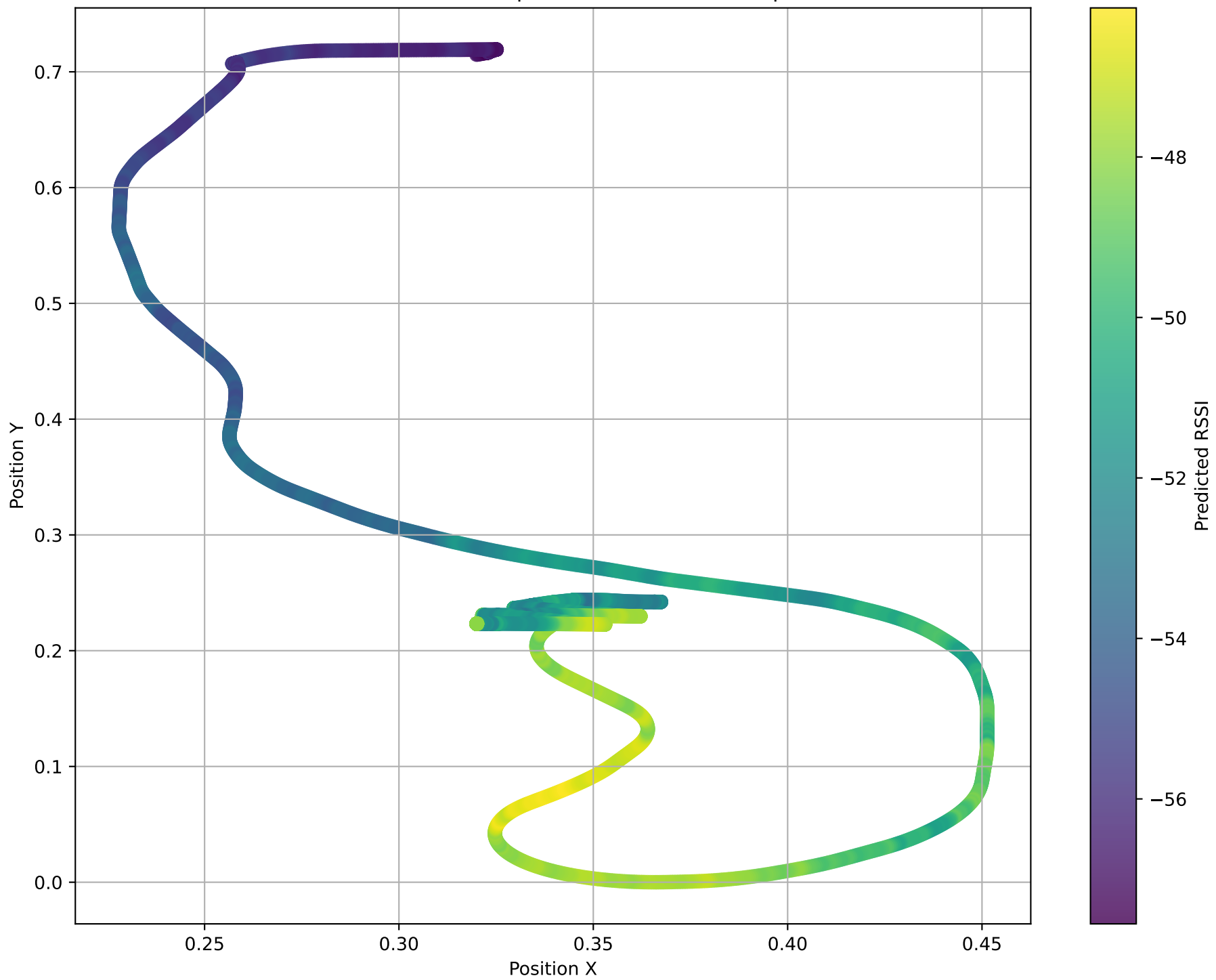
Distribution of Prediction Errors (First Step)



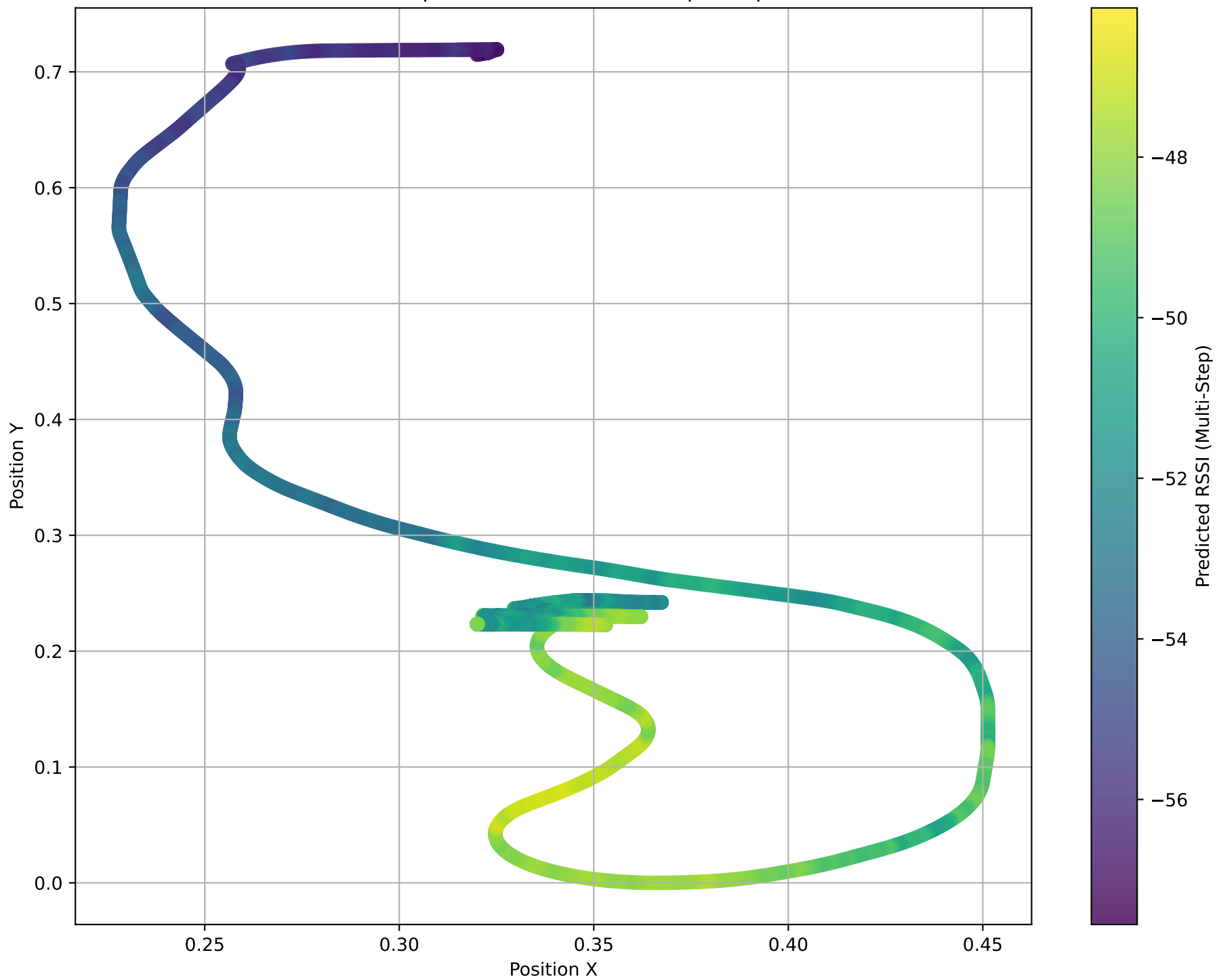
Distribution of Prediction Errors (Last Step - step 50 $\approx$ 2.0 s)



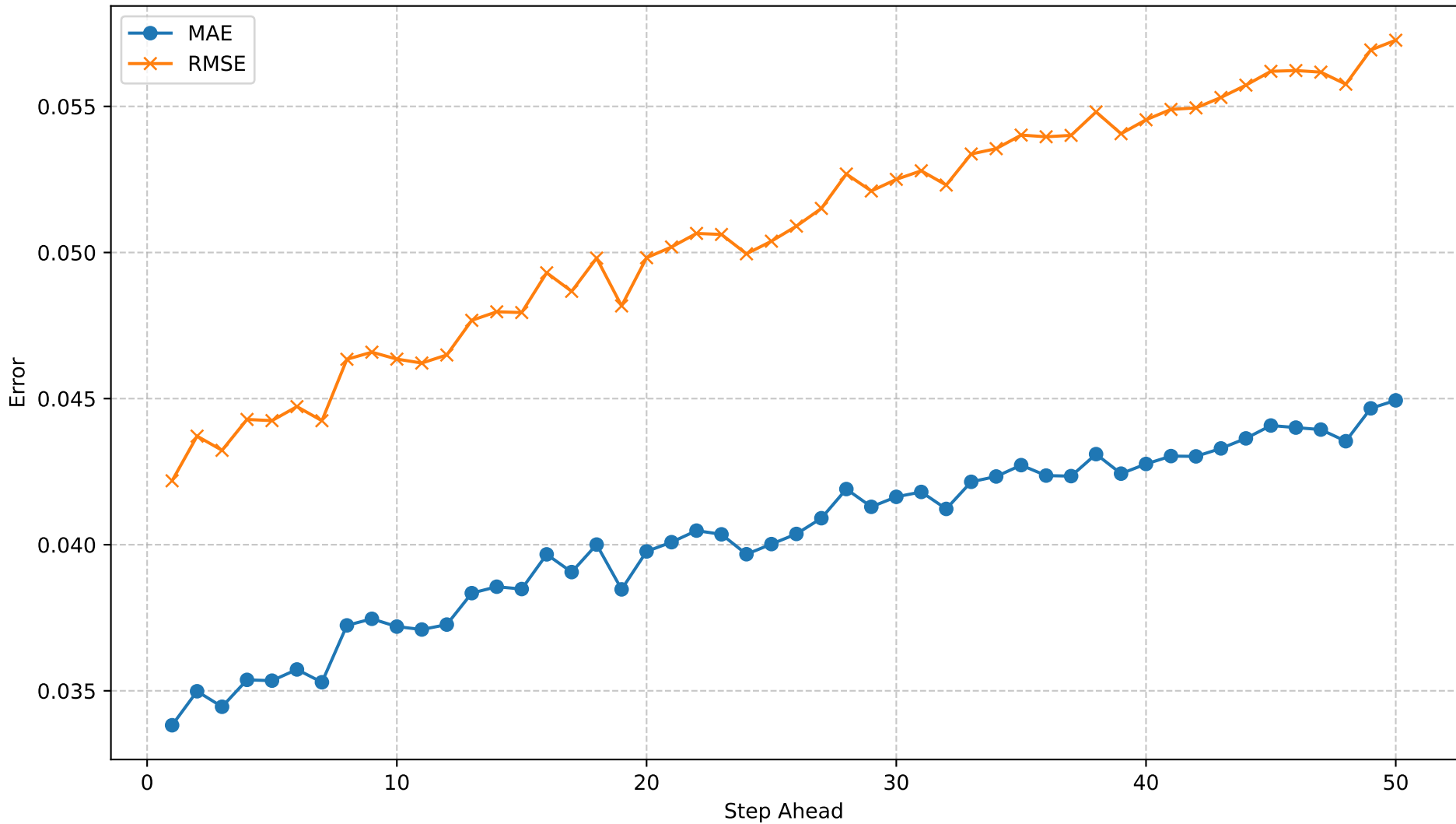
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 50 $\approx$ 2.0 s)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 50

MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.04 / 0.05
- 50º passo : 0.04 / 0.06

Crescimento do erro: 32.9% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 50 (0.04), 49 (0.04), 45 (0.04)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.